

Final Lab Exam (CPCS-204)

Total Time: 100 Minutes

Spring 2023

Max. Marks: 15

Name: _____ ID: _____ Section: _____

Note: Read the following instructions very carefully!

1. From **Blackboard**, download **Final Lab Exam**.
2. Save the .rar or .zip folder on **Desktop** or at your desired location.
3. Unzip this folder by right clicking on this folder.
4. Open **NetBeans** and select the option **File ➡ Open Project...** from the main menu.
5. Open all classes of your lab exam (4 in total) and **write your ID, Name** etc on top of every class and start writing the code in the classes given below.
6. Keep on saving your files after some time.
7. When you finish your exam, go to Blackboard and upload your solution after zipping the folder with the name (SectionName_yourID) which contains your solution.
8. **Only students are responsible for uploading the right solution folder to Blackboard. No invigilator is responsible to do so.**

You can add or delete any parameter in any class according to your requirements.

QueueLL class	Total	[4 marks]
Complete the functionality of the following methods:		
1. sumQueue() Method		
• Write code to FIND and RETURN sum of nodes of the Queue.		[1 mark]
2.addNodeAfter() Method		
• Write code to ADD NODE into the queue after a specific node .		[2 marks]
3.modifyAllNodes() Method		
• Write code to Modify the queue elements .		[1 marks]

FinalLabExam_Spring2023 class**Total [2 marks]**

Complete the functionality of the following method:

1. UpperTriSum () Method

Write code in this method to **Return** the **sum** of the upper triangle elements in a given nxn array (including the diagonal elements). You must use **recursion only**. DO NOT USE LOOP!

For example, the sum of the upper triangle including the diagonal elements of the following 3x3 matrix is **31**

1	9	4
6	5	10
5	4	2

- This method will be used in **choice 4** in the main class. Make the solution general to any nxn array.

FinalLabExam_Spring2023 class**Total [9 marks]**

- 1.** In **choice 1**, write code here to copy the elements of stack s1 into stack s2 such that the **largest element** in the stack s1 will be at the **top of the other stack (See sample below)**. Make sure your solution is general to any stack size.

Note:

[3 marks]

- (i) You can create other stack for this purpose (queue is not allowed).
- (ii) You should use **only** the methods defined in the stackLL class (push(), peek(),..etc).
- (iii) **Copy** the elements of stack s1 to stack s2 keeping in mind the largest of s1 is at the top of s2
- (iv) We do not care whether the second stack is ordered or not. We care only about the top to be the largest element in the second stack.

```

run:
|-----|
|----- Queue - Stack (Menu) -----|
|-----|
| 1. insert the largest element in the stack s1 into the top of stack s2      |
| 2. Add a node into the queue after a specific node                        |
| 3. Modify the queue elements with respect to the sum of the queue elements |
| 4. find the sum of the upper triangle (including diagonal) using recursion only
| 5. Quit                                                                    |
|-----|

> Please enter your choice: 1
S1:
35, 82, 90, 88,
S2:
90, 88, 82, 35,

```

2. In **choice 2**, write code here to **add node** into the queue **after a specific node** and then display all elements of the queue after adding this element. **[2 marks]**

- Ask the user after which node he wants to add a node and get this node.
- If this node does not exist, program should display the message: “The node n does not exist in the queue”.
- Otherwise, get Int value from the user to add into the queue q1.
- Add this value into the queue q1 (i.e., call addNodeAfter method)
- Display all elements of queue q1.

3. In **choice 3**, write code to **modify the queue elements** with **respect to sum queue elements** by performing the following actions. **[3 marks]**

- First check whether the sum elements of queue q1 is even or odd (i.e., call the above method (sumQueue()) then check the sum).
- If sum of the queue elements is **EVEN**, modify **all elements** of q1 by adding (+10) into each node (i.e., call the above method (modifyAllNodes())).
- If sum of the queue elements is **ODD**, modify **all elements** of q1 by subtracting (-10) into each node (i.e., call the above method (modifyAllNodes())).
- Display (Print) the elements of queue q1.

4. In **choice 4**, write code to call an appropriate method to find and display the sum of the upper triangle for a given nxn array (including the diagonal).

[1mark]